

Proxy-Resistant Benchmarks for Causally Grounded Finite Agents

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Abstract

Agent benchmarks often reward final task success, but final success can be proxy-supported: a model may exploit shortcuts, stale confidence, local final cues, uniform memory, or brittle tool surfaces. We propose a finite-agent benchmark design rule: a suite passes only when behavior, a suite-specific causal/structure gate, and anti-cheat controls all pass. We instantiate the rule with two hardened anchors. Suite C tests adaptive inquiry under world change: agents must remain quiet when stable, re-engage after change, recover attribution, avoid false calm, spend fewer probes than high-cost controls, and reopen after a second shift. Suite D/E tests long-horizon commitment: a future-critical variable must survive memory, generated action, tool schema, repair, causal readout, and black-box API behavior surfaces. The benchmark is not a consciousness test or production reliability certificate. It is a controlled diagnostic package for asking whether finite agents succeed for the right causal reason.

1 Introduction

Language-agent systems increasingly combine reasoning traces, external actions, memory, tools, search, and feedback [1, 9, 11, 13, 14]. End-to-end benchmarks evaluate whether these systems solve tasks in web, OS, software, and general-assistant settings [3, 6, 7, 12, 15]. These benchmarks are valuable, but final success alone cannot answer a different question: did the agent succeed because the future-relevant variable was represented, attributed, queried, and committed at the surface where action was selected?

This paper packages a family of finite diagnostic suites around one minimum pass rule:

A suite passes only if behavior passes, the relevant structure gate passes, and anti-cheat controls pass.

The rule is motivated by familiar safety and robustness failures. Reward hacking and specification gaming show that optimizing a measured score can satisfy the literal metric while missing the intended outcome [2]. Goal mis-generalization shows that agents can retain capabilities while pursuing the wrong objective out of distribution [4]. In benchmarks, the analogous failure is behavior-only success: the final answer is right, but the reason is wrong.

2 Benchmark Design

Each suite reports a vector score rather than a single leaderboard scalar: behavior, causal representation, attribution, inquiry, commitment, and generalization. The axis is included only when the suite can audit it.

Anti-cheat controls. Controls are suite-specific: matched distractors for memory, visible-control nulls for shortcut final cues, wrong-group transformations for OOD, shuffled source labels for attribution, signal suppression for false calm, matched-random probe budgets for inquiry, and repair/no-op controls for tool commitment.

3 Suite C: Re-Engagement Under World Change

Suite C tests whether an agent can reopen inquiry after its world model becomes stale. The finite harness contains two shifts. The first shift tests re-engagement after learned quiet; the second tests reopenability after a prior recovery.

The hand-policy terminal run passes all six gates. The headline `burst_then_refractory` condition reaches final affected MAE 0.112, first-shift selectivity 16.927, second-shift reopenability 16.667, and uses 22.6 probes. The matched scheduled and oracle references use 144.0 and 69.5 probes respectively. The false-calm control `fixed_surprise_decrement` looks quiet but fails recovery with final MAE 0.517.

The learned-transfer run trains a small probe head from teacher traces and evaluates it on held-out seeds. The learned head reaches final affected MAE 0.112, first-shift selectivity 16.667, second-shift reopenability 17.448, and uses 23.1 probes. Stale-signal, wrong-signal, signal-suppression, and matched-random controls fail in their intended ways.

4 Suite D/E: Commitment-Surface Memory

Suite D/E tests a temporal complement: an early variable matters when it controls a future action or tool surface. The benchmark moves the future-critical slot across runs and asks whether memory sensitivity follows that moved bottleneck rather than matched distractors.

Axis	Question	Example metric
Behavior	Does it solve the task?	final accuracy, return
Representation	Is the right variable tracked?	specificity, patch effect
Attribution	Is source assigned correctly?	self/world MAE
Inquiry	Does it ask when stale?	re-engagement, probe cost
Commitment	Does memory reach action?	tool field, repair value
Generalization	Does structure transfer?	compatibility, wrong group

Table 1: Vector score axes. The benchmark avoids collapsing distinct evidence types into one scalar.

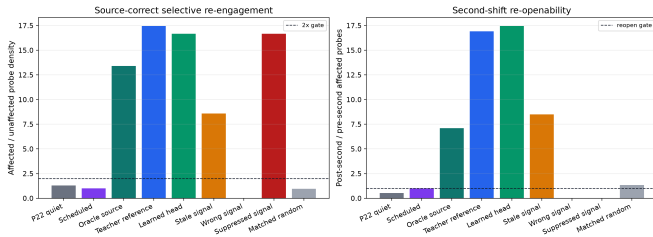


Figure 1: Suite C learned-transfer inquiry metrics. The learned head reopens after both shifts while matched random and wrong-source controls fail.

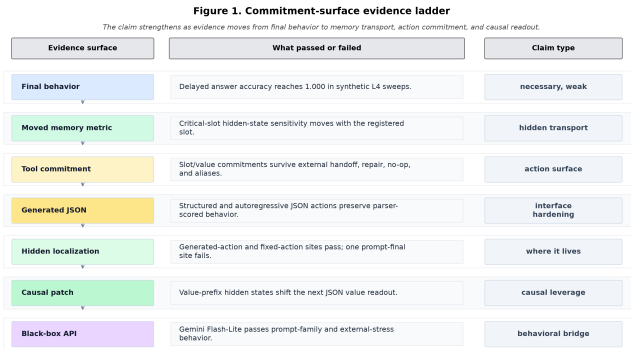


Figure 2: Commitment-surface evidence ladder. The claim strengthens when the future-critical variable survives behavior, hidden state, generated action, repair, causal readout, and API surfaces.

The evidence ladder includes synthetic trained agents, tool commitment, repair branches, generated JSON-like actions, prompt-level open-model behavior, hidden-state localization, fixed-action counterfactuals, causal patching, prompt-family robustness, and black-box API behavior. The strongest interpretation is not "the model remembers everything." It is the commitment-surface law: memory becomes agent-relevant when it survives the interface where it must control action.

5 Release Contract

A public benchmark release should include task generators, fixed seeds, declarative manifests, JSONL rows, summary JSON, benchmark cards, lower and upper baselines,

anti-cheat controls, and figure regeneration scripts. The shared schema requires a benchmark block, suite block, run metadata, minimum pass rule, score axes, gates, baselines, artifacts, allowed claim, and non-claims.

6 Related Work

The benchmark complements agent mechanisms such as ReAct, Reflexion, Voyager, SayCan, Toolformer, and Tree of Thoughts [1, 8, 9, 11, 13, 14]. It also complements end-to-end evaluations such as AgentBench, WebArena, OSWorld, GAIA, SWE-bench, HELM, and BIG-bench [3, 5–7, 10, 12, 15]. Our contribution is narrower: finite diagnostic suites that reveal whether success is supported by the intended causal structure.

7 Limitations

The current suites are finite and controlled. Suite C’s learned head is teacher-trained, not reward-trained or open-agent learned. Suite D/E black-box API runs are behavioral and do not establish hidden-state localization for closed models. Synthetic JSON/tool ladders do not certify autonomous natural-language tool reliability. The benchmark does not test consciousness, human cognition, biological agency, or production safety.

8 Conclusion

Proxy-resistant benchmarking changes the pass condition. Final behavior is necessary but not sufficient. A finite agent should pass only when the future-relevant structure reaches the prediction, attribution, inquiry, memory, or commitment surface where action is chosen, and when anti-cheat controls rule out the shortcut.

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A Suite Inventory

The full benchmark family has six suites:

- Suite A: consequence-to-action.
- Suite B: reafferent source attribution.
- Suite C: re-engagement under world change.
- Suite D: long-horizon moved bottleneck.
- Suite E: tool commitment and repair.
- Suite F: structure-compatible OOD.

The main paper centers Suite C and Suite D/E because they are the strongest current anchors. Suites A/B/F should remain in appendix or a suite-family overview unless expanded into their own benchmark cards.

B Suite C Gate Definitions

Source: `experiments/world_responds/results/suite_c_reengagement_2026_07_06.md` and `experiments/world_responds/results/suite_c_neural_transfer_2026_07_06.md`.

Gate	Meaning	Current status
C1	baseline silence replication	pass
C2	selective first-shift re-engagement	pass
C3	affected-component recovery	pass
C4	no false calm	pass
C5	cost-aware inquiry	pass
C6	second-shift reopenability	pass
N1	stale/wrong/suppressed learned controls fail	pass in learned transfer

Table 2: Suite C gates.

C Suite D/E Gate Surface

Source: `experiments/long_horizon_bottleneck/BENCHMARK_CARD.md`.

The benchmark should report:

- closed-loop final accuracy,
- moved-slot specificity and rank,
- visible-control null behavior,
- hidden localization over model/layer/token-position grids,
- fixed-action counterfactual localization,
- causal patch value-prefix effects,
- prompt-family robustness,
- API black-box prompt-family and external-stress behavior.

D Release Schema

The machine-readable summary must contain:

```
benchmark, suite, run, minimum_pass_rule,  
score_axes, gates, baselines, artifacts,  
allowed_claim, non_claims
```

Rows should be JSONL. Summary JSON should be public-safe and should not include secrets, uncured provider transcripts, or local-only Modal payloads.

E Known Negatives and Boundaries

Known negatives are valuable and should not be hidden:

- signal-layer cooling can create false calm in Suite C,
- matched-random inquiry at the same budget fails selectivity/recovery,
- stale and wrong stress signals fail the learned Suite C transfer,
- API dispatch failure for one tested OpenAI condition is sparse and localized, not broad,
- Suite F remains a companion OOD package rather than a complete benchmark release inside this paper.